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Quantifying the value of carbon label information in food choice using drift diffusion modelling

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ABSTRACT

The use of carbon labels as an intervention to increase more sustainable food consumption has seen many mixed results, with some studies showing that consumers do not utilise the carbon labels in their decisions. To address the mixed results in the literature, we present a novel and in-depth evaluation of how carbon labels work by quantifying the importance of carbon label information relative to taste preferences in food decisions via a computational modelling approach. Participants (n = 48) were presented with multiple trials of two sandwiches alongside their carbon labels. Participants' choice and response time were recorded whilst visual attention was tracked with an eye-tracking device. The Multi-attribute Attentional Drift Diffusion Model (maaDDM) was fitted to data through Bayesian STAN modelling in R. The analysis revealed that carbon labels were used to a moderate extent similar to individual taste preference in choosing sandwiches, but the extent of use varied as a function of participant's perception of the negative impact of GHG emissions (the more negative perception, the greater use of carbon labels). We further explore the insights gained from maaDDM on participant's information sampling behaviour, and discuss the implications for policies to identify a critical valuation threshold of carbon labels.

1. Introduction

According to reports from the Intergovernmental Panel on Climate Change (IPCC, 2018), our global food system poses one of the biggest threats to worsening climate change in the Anthropocene. Indeed, recent studies have found that our global food consumption alone contributed to around 36% of worldwide greenhouse gas (GHG) emissions (Xu et al., 2021), with ever-increasing levels of meat consumption continuing to fuel these high figures (Gerber et al., 2013; Mattick et al., 2015). As such, immediate and effective efforts to reduce our meat consumption is crucial in our fight against climate change (Kwasny et al., 2022). In fact, the individual action with the greatest potential to reduce GHG emissions was shown to be through eating less meat (Lacroix, 2018). To achieve this, consumption-based behavioural interventions (i.e., interventions targeting to change levels of consumption) may be most effective in encouraging consumers to transition towards a more sustainable diet with less meat (Londokova et al., 2021). For example, the use of carbon footprint labels (which conveys information on carbon emission) may encourage low-carbon food consumption and discourage meat-eating (Liu et al., 2016). However, the literature has seen mixed results regarding the use of carbon labels to increase more sustainable food choices (Kwasny et al., 2022).

In this paper, the use of carbon labels is re-evaluated within an empirically-grounded framework. We explored the underlying

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cognitive processes of how carbon label information is sampled, valued, and weighted in food decisions through the use of computational modelling and eye-tracking methods. We demonstrate a novel quantification of how carbon labels were valued and explore the variability in individual preference for using carbon labels. Finally, we discuss and argue that carbon labels may still be an effective intervention, and conclude its implications for policies.

1.1. Literature review

In the domain of food consumption, downstream interventions (e.g., eco-labelling, framing, and social norms) which focus on manipulating the type of food product information provided to the consumers (Lehner et al., 2016) have gained widespread popularity within academic spheres and policy recommendations due to their low-cost and easy implementations (Brazil and Caulfield, 2017). One such example is the use of carbon footprint labels (hereinafter carbon labels). Also known as eco-labelling, this intervention aims to inform consumers of the amount of carbon footprint (i.e., a standardised measure of GHG emission; Brander and Davis, 2012) emitted to produce a product through a sticker or label on the packaging (Lemken et al., 2021). Carbon labels explicitly display information on the environmental sustainability of a product which is not readily observable, thus giving consumers access to more information regarding the product to facilitate more comparisons before choosing (Cohen and Vandenbergh, 2012). This approach is expected to raise consumer's awareness of the environmental impact of their food consumption (Camilleri et al., 2019), encouraging more sustainable consumption choices (Lucas et al., 2008).

However, studies on the use of carbon labels tend to yield relatively more mixed results (Kwasny et al., 2022; Nisa et al., 2019). For instance, the presence of carbon footprint labels has been found to lead to more sustainable food choices (Brunner et al., 2018). Online experimental studies have shown that consumers make more sustainable consumption choices when carbon labels are located on hypothetical restaurant food menus (Betz et al., 2022), or located on coffee products at point-of-sale (Thøgersen and Nielsen, 2016; see also Meyerding et al., 2019; Visschers and Siegrist, 2015; Vlaeminck et al., 2014). Carbon labels were also found to be effective in field studies, showing that consumers purchased 20% more low-GHG-emission products when carbon labels were placed on 37 grocery items (e.g., milk, canned tomatoes) over 3 months (Vanclay et al., 2011).

On the other hand, other studies found that carbon labels had little-to-zero effect in increasing more sustainable food choices, thus suggesting that carbon labels are not very effective as an intervention (Babakhani et al., 2020; Lin and Nayga Jr, 2022). The use of carbon labels did not increase more sustainable choices in Lin and Nayga Jr's (2022) study, whereas carbon labels were found to only increase sustainable food choices by a small amount of 2.7% in Lohmann et al.'s (2022) study. These findings are further supported by a meta-analysis from Nisa et al. (2019), demonstrating that the overall effectiveness of carbon labels in promoting sustainable choices accumulated to only a small degree of 3.4%. Compared to other interventions such as choice architecture or nudges (24.9% effectiveness) and goal-oriented engagement strategies (17.9% effectiveness), this suggests that the use of carbon labels was not very effective in encouraging sustainable behaviours (see also Kwasny et al., 2022, for a systematic review). In summary, the mixed evidence from the literature suggests that carbon labels may not be as effective as desired given their small effect size.

To address the mixed results in the literature, we diverged from the typical experimental A/B-testing of whether presence (vs. absence) of carbon labels reduces carbon footprint in food choices as seen in studies above. Instead, we quantified the extent to which sustainability as a factor is traded off against personal taste preferences when carbon labels are present, which is one of the strongest and most habitual factors in food decision (e.g., Boesveldt et al., 2018; Grunert et al., 2014). In the following section, we explore the psychological mechanism behind how carbon labels work and the issue in measuring importance, and then describe the computational modelling and eye-tracking approach in detail.

1.2. Using carbon labels in food choices

According to Grunert's (2011) and Grunert et al.'s (2014) theoretical model, the psychological mechanism behind effective use of carbon label can be broken down into three parts: *attention*, *knowledge*, and *importance*. This model suggests that consumers need to (i) pay attention to the carbon label, (ii) understand what it means and how to use it, and (iii) place enough importance to it for the carbon label to be used effectively in guiding their decision making (Juhl and Poulsen, 2002, in Grunert, 2011). This study focuses on the third component—*importance*.

Previous studies have found evidence for the model above (Grunert, 2011). In Babakhani et al.'s (2020) eye-tracking study, multiple burger options (e.g., meat and vegetarian) were presented alongside carbon labels indicating the amount of carbon footprint produced by each burger. Participants were instructed to choose a burger of their preferred choice whilst the eye-tracking device recorded their visual gaze location and looking time. Results showed that, on average, participants spent only 5% of their total looking time on carbon labels (i.e., only 2.63s out of 56.47s), whilst 40% of participants took a short glance of less than 1s at the labels. Furthermore, 79% of participants in Babakhani et al.'s (2020) study reported not understanding what the labels meant and how they work, potentially explaining why they paid little attention to the labels. Consequently, they did not give much importance to the labels which were "not interesting enough" (p. 59) to have influenced their burger decision-making (Babakhani et al., 2020). Indeed, compared to a control group without any labels, the presence of carbon labels did not significantly reduce the number of meat burgers chosen. Overall, this suggests that participants paid relatively less attention to the carbon labels and thus did not use them at all during their food decision-making, which may go some way to explain the lack of effective use of carbon labels to inform sustainable decision making in some of the previous studies (e.g., Lin and Nayga Jr, 2022).

Another study (Dudinskaya et al., 2020) in the line of econometric modelling examined the use of carbon labels (present vs. absent; amongst multiple other attributes such as price and protein content) in a discrete choice experiment (DCE) looking at meat purchase

behaviour, where they focused on the impact of attribute non-attendance (i.e., not taking an attribute into consideration) using participant self-reports and eye-tracking. Results showed that carbon labels were the second least attended to attribute according to participant self-reports, and the least looked at attribute according to eye-tracking data. Consequently, the presence of carbon label was not a significant attribute in participant's decision-making regardless of whether it was attended to or not (both visually and self-reported), thus suggesting no significant contribution to decision utility and no influence on decision (see also Edenbrandt et al., 2022, for similar results looking at the alternative or option counterpart of non-attendance). Moreover, participants in the study were also not familiar with the carbon labels and how it works, thus reinforcing the notion of (lack of) knowledge and attention as key barriers in the use of carbon labels.

Overall, whilst these findings highlight the need to tackle the barriers of attention and knowledge, they also put forward two important questions in the literature: (1) How much importance would consumers give to carbon labels and use them in their food decision when aided in attention and knowledge as desired in eco-labelling intervention? (2) How can we measure the extent of this importance and use of carbon labels?

In previous studies, the importance given to carbon labels was mainly measured through self-reports (Grunert et al., 2014). For instance, participants were asked to rank the importance of sustainability amongst other factors, such as taste preference, health considerations, price, and branding, in their food decision making (Babakhani et al., 2020), or to rate the extent to which they used the carbon labels in their choice as measure of importance (i.e., the greater the importance of carbon label, the more likely it will be used to guide a decision; Grunert et al., 2014). In most cases, participants' food choices would be motivated more by their personal taste preferences (i.e., whether they like the taste of the product) than sustainable considerations (Grunert et al., 2014; Roos and Tjarnemo, 2011; see also Embling et al., 2024; van den Berg et al., 2022), but self-reported measures and revealed preferences are limited in conveying the extent to which taste preferences and sustainability each contribute towards the final decision. For example, when consumers rank taste preferences as more important than sustainability, the possibility of division in contribution could range from a distinct contrast of 99 % taste and 1 % sustainability (ratio = 99), to an almost even split of 51 % taste and 49 % sustainability (ratio = 1.04). This further relates to the idea of a critical threshold: the extent to which carbon labels need to be valued to influence a decision to become more sustainable (e.g., choosing a more sustainable food option over a less sustainable one). Hence, efforts to examine where the division lies on the scale of importance between taste preferences and sustainability may provide a novel yet crucial perspective to evaluating the effectiveness of carbon labels. This could further shed light on the extent to which sustainability is traded off with other factors in food decision such as price and health considerations in future studies. This study addresses this gap in knowledge.

As carbon labels work by consumers using the carbon footprint information provided, it is proposed that an information sampling model may be most appropriate to study how much carbon label information is valued and used in decision-making. Drawing from the field of computational modelling, the extent to which consumers value sustainability and use carbon labels can be measured using the Multi-Attribute Attentional Drift Diffusion Model (maaDDM), developed and validated by Yang and Krajbich (2023). This model suggests that, when given two choices with two attributes (i.e., qualities of a product, such as taste vs. sustainability), visual gaze attention influences the weightage (i.e., importance) of each attribute during decision-making (Diederich, 2003). For instance, longer gaze at one attribute (e.g., taste) leads to stronger discounting of the other attribute (e.g., sustainability), which in turn promotes choices in favour of the former attribute (e.g., taste over sustainability). In the context of sustainability research, the maaDDM assumes that if consumers attribute a low value to the carbon footprint information, they will allocate less gaze and attention to the carbon labels, thus giving less weightage to the carbon labels and less consideration for sustainability to guide their food decision. Critically, the maaDDM can generate a precise and accurate quantitative measure (between 0 and 1) for the importance given to carbon footprint information conveyed by the carbon labels (relative to another attribute such as taste preferences), which goes beyond the simpler measures of self-reported measures and revealed preferences to measure importance.

Hence, in this study, we aimed to explore the quantitative extent to which sustainability information provided by the carbon labels is valued and used in food decisions as compared to personal taste preferences using the maaDDM. Whilst we expected taste preferences to be weighted as more important than sustainability during food decisions, the study remained exploratory given there are no prior results to draw from or compare with. Furthermore, we examined the individual variability in preferences for using carbon label information in an exploratory analysis to understand which climate attitudes predict greater sustainability considerations in food choices. The novel maaDDM (Yang and Krajbich, 2023) was applied via the combination of computational modelling and eye-tracking methods, whilst the barriers of attention and knowledge (Grunert et al., 2014) was accounted for in a food choice task.

2. Method

2.1. Participants

A total of $n = 78$ participants signed up for the study (female = 58, male = 18, non-binary = 2; $M_{age} = 20.32$, $SD_{age} = 2.48$; British = 37, non-British = 41), but 30 of them were excluded from analysis (see 3.1 Data Cleaning for reasons). Participants were mainly psychology undergraduates recruited through the University of Bath's Research Participation Scheme (RPS; $n = 58$). The remaining included university students recruited through word-of-mouth. As incentive for participation, undergraduates who signed up through RPS received course credits for academic purposes, whilst non-RPS participants had a chance to win one of five £20 Amazon vouchers through a raffle.

2.2. Materials

2.2.1. Carbon labels

Carbon labels can be found in various shapes, colours and sizes, each design conveying different information to the viewer (see review by Rondoni and Grasso, 2021). In this study, we re-designed the carbon labels according to good practices and recommendations from the literature. Firstly, the carbon labels should follow a traffic-light system which is familiar to most consumers, where green, yellow and red signify the relative carbon footprint emissions as low, medium or high, respectively (Thøgersen and Nielsen, 2016). Secondly, showing the absolute carbon footprint (in the format of CO_2-equivalent, or CO_2-e) helps provide a standardized metric to measure and convey the environmental impact of a product (Lemken et al., 2021). Thirdly, the labels should incorporate a grading system to help put the carbon footprint emissions into context and ease comparison between products (Lohmann et al., 2022). Taken together, three label designs (*Bar*, *Letter*, and *Scale*) were generated according to these guidelines (Fig. 1A).

The labels were tested in a within-subjects pilot study using a different sample ($n = 30$) for its quality performance, which included clarity, ease of understanding and reading, overall familiarity, and its popularity ranking as the most preferred label. In general, all labels were rated as clear, easy to read and understand, familiar and easily associated as carbon footprint labels (see Table 1; most means > 4.0). The measurements were combined (except popularity ranking) and averaged to give a Label Performance score. Repeated-measures ANOVAs (Norman, 2010) revealed that there were no significant differences between all three labels for the Label Performance score, $F(2,58) = 1.79$, $p = .177$ (contrasts: all p's $\geq .257$), and Popularity Ranking, $F(2,58) = 2.11$, $p = .131$ (contrasts: all p's $\geq .252$), indicating that the labels were similar in their quality performance and popularity.

Nonetheless, the Scale-type design was chosen for this study as it was the most popular design (i.e., highest absolute ranking) and had received good comments such as "design is nice and easily understood". The Scale-type design was then split into five categories, ranging from A (*least carbon footprint produced*) to E (*most carbon footprint produced*). The categories were colour-coded by the traffic-light system accordingly to show that green is good whilst red is bad for the environment. Each carbon label also has its own numerical number to indicate the amount of absolute carbon footprint produced in terms of CO_2-equivalent rate in grams (Fig. 1B). Each carbon label category was also given a Carbon Score (from A = 5 to E = 1).

2.2.2. Knowledge check

To ensure all participants have the same understanding of the carbon label system (hence checking the *knowledge* barrier), infographics were created to introduce and explain the carbon label system (Fig. 2). Of utmost importance, participants must understand the following system: (a) the conversion unit of "grams of CO_2-e" indicates the amount of greenhouse gas produced equivalent to releasing X-grams of carbon dioxide (CO_2) to the environment (where X can be any number); (b) an alphabetical order is used to categorise the most sustainable options as "A" (least carbon footprint produced) and the least sustainable option as "E" (most carbon footprint produced); (c) the categorization above is comparable only within each food type (e.g., meat vs. meat), and cannot be compared across different food type (e.g., meat vs. milk is not allowed).

A quick knowledge check questionnaire was also administered to ensure all participants understood the carbon label system correctly. The questionnaire consisted of four questions covering all fundamental knowledge above (Supplementary Material A). Participants who got any question wrong were not penalized but instead given a full explanation of each mistake by the researcher before moving on with the study.

2.2.3. Food stimuli

Sandwiches were chosen as the food stimuli for this study for a few reasons. Firstly, sandwiches play a huge role in the everyday lives on UK citizens. According to the British Sandwich and Food to Go Association (BSA), UK citizens consume around 11.15 billion sandwiches each year (often bought as part of a meal deal for lunch), generating roughly £7.85 billion in revenue (BSA, 2017). Hence, sandwiches should be a common type of food familiar to most people. Secondly, there are multiple types of sandwiches to choose from (e.g., Ham and Cheese, Prawn Mayo, BLT, etc.), which also covers a wide range of carbon footprint produced. As such, sandwiches were deemed very suitable for the current study as it allows the required within-food type comparison to be made easily. Lastly, the

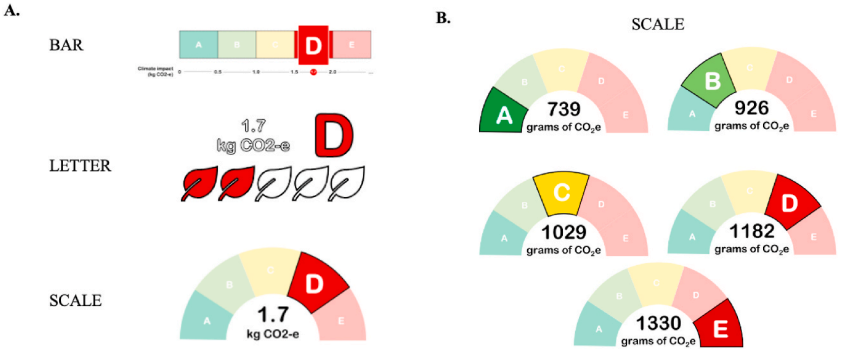


Fig. 1. Types of carbon label design.

Table 1
Descriptive statistics for label performance and popularity.

	Label Type					
	Bar		Letter		Scale	
Measures	***M***	***SD***	***M***	***SD***	***M***	***SD***
Performance						
1. It was easy for me to understand what these labels mean	5.63	1.35	5.43	1.61	5.70	1.26
2. I am familiar with this type of label design	4.07	1.89	3.67	1.99	4.50	2.11
3. The label was clear in conveying its meaning	5.60	1.50	5.20	1.35	5.57	1.38
4. It was easy to read the labels	5.50	1.66	5.07	1.70	5.80	1.19
5. It was easy to associate these labels as carbon footprint labels	5.37	1.52	4.50	1.85	4.90	1.54
Total Average	5.23	1.32	4.77	1.43	5.29	1.13
Popularity						
1. Which label design did you like the most?	2.17	0.84	2.13	0.78	1.70	0.79

Note. Scale for Performance: 1 = *strongly disagree*, 7 = *strongly agree*. Scale for Popularity: 1 = *like the most*, 3 = *like the least*.

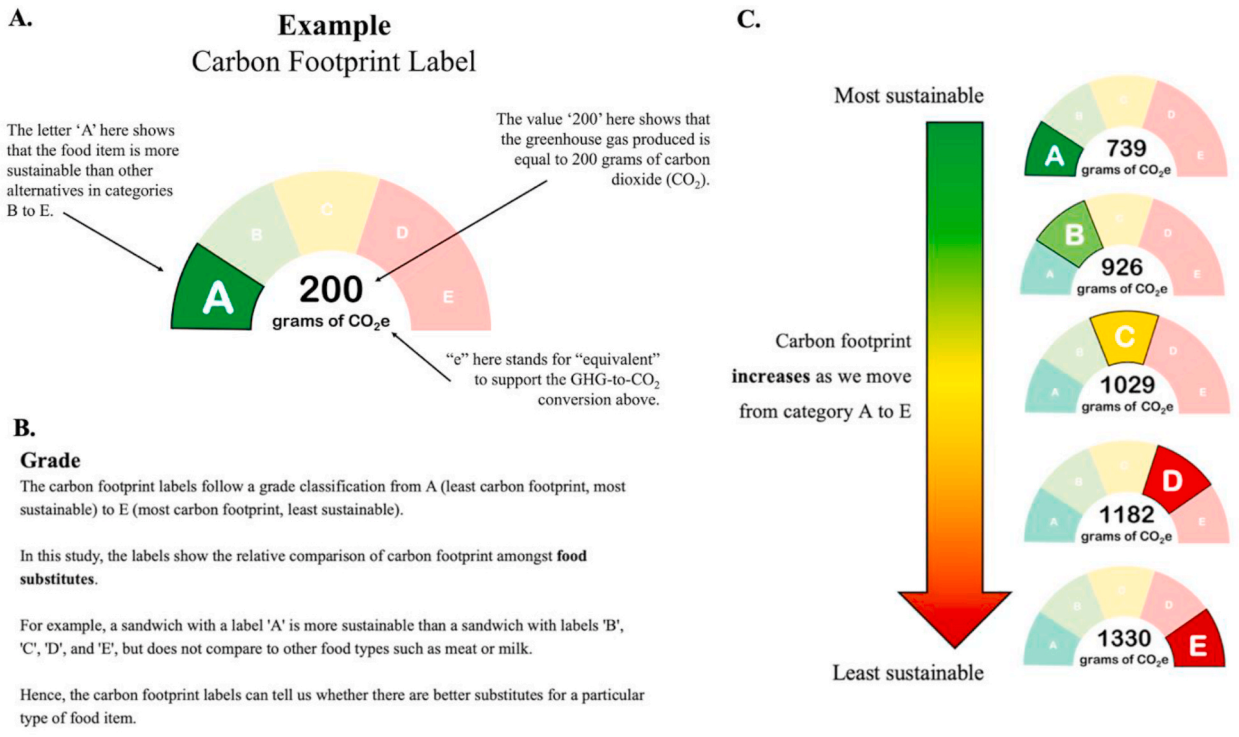


Fig. 2. Infographics on carbon label system.

carbon footprint information of sandwiches has been well-studied and readily supplied in the literature by Espinoza-Orias and Azapagic (2018). This provides an informed guide on carbon footprint produced by each pack of sandwich, thus making the comparisons between sandwiches more valid. Overall, 15 different sandwiches of a wide range of carbon footprint were selected. Images of sandwiches were taken from Sainsbury's website (https://www.sainsburys.co.uk), although some sandwich names may differ from Espinoza-Orias and Azapagic's (2018) list of sandwiches (e.g., "Cheese and Ploughmans" was replaced by "Red Leicester Ploughmans").

As seen in Table 2, the sandwiches were divided into five categories (three sandwiches within each) from high to low carbon footprint, with the category division following the equation of (max–min)/5 = (1444.3–739.0)/5 = 141.06 as the range within each category. The sandwiches were then assigned their carbon labels (from A to E), each with their unique numeric quantity of carbon footprint produced in the labels. All food stimuli were assumed to have the same price, replicating a 'meal deal' (a main, a snack and a drink combination popular in the UK) which costs the same regardless of sandwiches (main) chosen.

2.2.4. Taste preferences

Participants were asked to self-report their taste preferences from 1 (*very dislike*) to 5 (*very like*) on an analogue-sliding scale (i.e., values can fall in between integers) for all 15 sandwiches on Qualtrics. A higher score indicated a stronger preference or liking for the

Table 2
Carbon footprint and categorization of sandwiches.

Sandwich Name	Carbon Footprint (grams of CO_2-e)	Label Category
All Day Breakfast	1444.3	E
Smoked Ham and Cheddar	1349.5	E
Smoked Salmon and Cream Cheese^	1330.0^	E
Prawn Mayo	1254.7	D
Tuna Mayo^	1213.0^	D
Bacon and Free Range Eggs	1182.4	D
Red Leicester Ploughmans	1112.1	C
Cheese and Red Onion	1078.4	C
Chicken and Bacon	1029.7	C
BLT (Bacon, Lettuce and Tomato)	1007.4	B
Lighter Tuna and Cucumber	1000.5	B
Chicken with Pork, Sage and Onion Stuffing	926.9	B
Tuna and Sweetcorn	851.6	A
Chicken and Sweetcorn	769.4	A
Free Range Egg and Cress	739.0	A

Note. Best estimates of carbon footprint were used for sandwiches with ^ mark as absolute value not given in Espinoza-Orias and Azapagic (2018). The categorization follows the range of 739.0–880.06 for A, 880.06–1021.12 for B, 1021.12–1162.18 for C, 1162.18–1303.24 for D, and 1303.24–1444.3 for E.

sandwich.

2.2.5. Food choice task

A two-choice (Left and Right), two-attributes (Taste and Sustainability) food choice task was built in PsychoPy version 2022.2.5 (Peirce et al., 2019) as the main task for the study. As seen in Fig. 3, two sandwiches (proxy for Taste) were presented on both upper left and upper right sides, whilst the carbon label (proxy for Sustainability) specific to each sandwich was presented on both lower left and lower right sides of the screen. All 15 sandwiches with their specific carbon labels were paired against one another for comparison, thus creating a total of 105 unique combinations, or trials (in mathematical terms, $^{15}C_2 = 105$). All 105 trials were randomly divided into five blocks (21 trials in each block), and each block was presented in a random order to the participants.

During the task, participants saw a fixation cross in the middle of the screen (for 1s) before each trial and were then presented the comparison trial (Fig. 3) for up to 20s. Participants were then required to compare both sandwiches and choose the one sandwich (Left or Right) that they prefer more than the other by pressing "Z" for the Left option and "M" for the Right option. Each trial ended once a decision was made, or 20s had elapsed (7 out of 8 190 trials), and participant's choice (Left = 1, Right = 0) and Reaction Time (in seconds) were recorded. After each block (21 trials), participants were given a break for as long as they wanted (at least 5s) before continuing to the next block. Technical details of the creation of food choice task in PsychoPy are reported in the Supplementary Material B.

2.2.6. Eye-tracking

Participant's visual gaze attention was tracked using an EyeLink 1000 Plus at a sampling of rate of 1000Hz throughout the food choice task. During the task, participant's gaze dwell time, which included both saccades (20–40ms) and fixations (200–300ms;

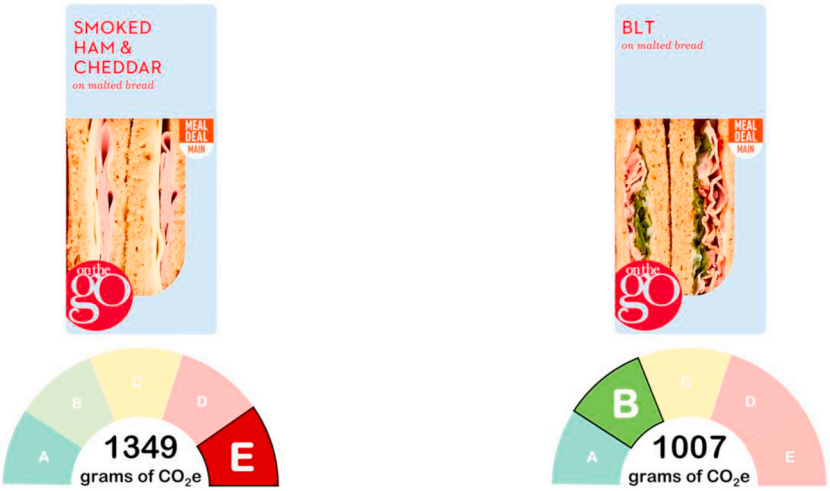


Fig. 3. Example trial in food choice task.

Duchowski, 2007), was recorded (in seconds) for all four quadrants of stimuli (i.e., sandwiches on upper left and upper right, and carbon labels on lower left and lower right). A 13-point calibration was conducted before the experimental task. Gaze deviation threshold was set to one degree, as deviations within one degree was considered normal and good without any significant error (SR Research Ltd, n.d.). Where calibration failed, we attempted to make adjustments (e.g., changing height of chin rest, camera angle, illuminator intensity, or screen height, etc.) until calibration is passed. Whilst this solved most calibration issues, we note that some participants ($n = 12$) still could not pass calibration despite all the adjustments made. Nonetheless, they were still allowed to complete the food choice task, but were removed from analysis (see 3.1 Data Cleaning below). A chin rest was equipped and situated 60cm from the screen.

2.2.7. Climate attitude measures

Participant's perception and attitudes towards climate change were measured across multiple scales on Qualtrics. Each scale is described below.

i. **Threat.** Participant's level of perceived threat of climate change was measured on a 6-item, 5-point Likert scale (Cronbach's alpha = 0.72) adapted from Whitmarsh and Corner (2017). The scale asked participants how serious of a threat they think climate change poses towards different actors (e.g., them and their family, wildlife and ecosystem, etc.). Participants responded from 1 (*not serious at all*) to 5 (*extremely serious*); a higher average score indicated a higher level of perceived threat of climate change overall.
ii. **Willingness.** Participant's willingness to adopt a "green" diet was measured on a 4-item, 7-point Likert scale (Cronbach's alpha = 0.80) adapted from Capstick et al. (2015). The scale asked questions regarding participant's willingness to change their eating habits, such as "Cut down the amount of meat consumed" and "Choose a vegetarian or vegan meal when eating out", and responses ranged from 1 (*not at all willing*) to 7 (*extremely willing*). A higher average score indicated higher willingness to change to a "green" diet.
iii. **NegativeGHG.** A 2-item, 7-point scale (Cronbach's alpha = 0.46) measured the extent to which participants envision the negative impact of climate change will be for "people their age" and "the future generations" if we do not reduce GHG emissions soon (adapted from Brugger et al., 2020). Responses ranged from 1 (*extremely small*) to 7 (*extremely large*), with a higher average score indicating larger perceived negative impact of climate change.
iv. **Impact.** A 1-item, 7-point Likert scale (adapted from Poortinga et al., 2018) measured the extent to which participant thinks the impact of climate change will be good or bad for people across the world. Responses ranged from 1 (*extremely bad*) to 7 (*extremely good*). A lower score indicated a stronger perception that the impact of climate change will be bad for everyone.
v. **PolicySupport.** Participant's level of support towards climate policy was measured with a 2-item, 5-point Likert scale (Cronbach's alpha = 0.33) adapted from Whitmarsh and Corner (2017). Participants were asked to what extent they support or oppose the following policies: "The UK government signing up to an international agreement to tackle climate change" and "People in the UK making changes to their lifestyles to reduce their impact on the environment", on a scale of 1 (*strongly oppose*) to 5 (*strongly support*). A higher average score indicated higher levels of support towards climate policies.

2.3. Procedure

Participants were invited to the eye-tracking lab, were given the information sheet and provided their informed consent to participate. Participants then learned about the carbon label system using Qualtrics. A knowledge check was administered to ensure all participants understood how the carbon label system works. Participants were then moved to a different computer and sat in front of the eye-tracking device for calibration. Next, participants were given instructions on the food choice task before completing the task on PsychoPy whilst gaze attention was measured via eye-tracking. Upon completion, participants were moved back to Qualtrics to provide their self-reported taste preferences for each food item. Lastly, participants answered several questions on climate change perception and attitudes, and provided their demographic information (e.g., age, gender, nationality, and any dietary restriction) before being given a debrief at the end of the experiment. The whole study typically took 30–40min to complete. This study received ethical approval from the University of Bath Psychology Research Ethics Committee (PREC code: UGM 23-013).

2.4. Model Specification

The framework for the design and analysis of this study follows the Multi-Attribute Attentional Drift Diffusion Model (maaDDM), which was recently developed and validated by Yang and Krajbich (2023). The maaDDM—an extension of the basic DDM—is based on the sequential sampling framework which conceptualizes the decision-making process as an accumulation of evidence or information over time, and a decision is made when the information (favouring one choice relative to another) reaches a decision boundary (Ratcliff, 1978). In the context of food decision-making, the sequential sampling framework assumes that consumers sample information from product qualities (e.g., its ingredients, costs, health benefits, environmental impact, etc.) until enough information is accumulated (e.g., knowing the product is tasty, cheap, healthy, and sustainably produced) to reach a decision boundary (e.g., to buy this product or not). As such, the sampling of evidence or information is always relative in nature between two or more options and attributes within the maaDDM, meaning that the different options and attributes are sampled in competition to favour one choice over another.

The maaDDM is built upon two models of value-based decision making, namely the Decision Field Theory (DFT; see Diederich,

2003) and Attentional Drift Diffusion Model (aDDM; see Gluth et al., 2020; Krajbich and Rangel, 2011). In the DFT, gaze attention is assumed to fluctuate between different attributes (i.e., qualities), thus changing the weights on these attributes in the evidence accumulation process. The more gaze attention given to one attribute over another (e.g., price vs. health benefits of a product), the more important the evidence sampled from that attribute. Furthermore, the aDDM assumes that gaze attention fluctuates between options (e.g., product A vs. product B), and the more gaze attention given to one option over the other would thus favour the evidence coming from that option. Taken together, Yang and Krajbich's (2023) maaDDM merges both ideas and assumes that gaze attention fluctuates between options and attributes, thus amplifying the weight of evidence depending on which option and which attribute is paid attention to.

The model building for maaDDM above is explained in simpler terms as follows (see Yang and Krajbich, 2023). Let us assume that there are two choices $W \in \{X, Y\}$ with two attributes $i \in \{A, B\}$. This would result in four elements $W_i \in \{X_A, Y_A, X_B, Y_B\}$, with each element W_i representing option W with attribute i. According to the aDDM (which describes option-wise attention), the evidence accumulation rate (i.e., drift rate) when comparing between options would be:

$$(X_A - \theta Y_A) + (X_B - \theta Y_B) \textit{when looking at option } X,$$

$$(\theta X_A - Y_A) + (\theta X_B - Y_B) \textit{ when looking at option } Y,$$

where θ represents the discount factor on the non-fixated (i.e., not looked at) option ($0 \leq \theta \leq 1$). When option X is looked at, for example, evidence is accumulated from both attribute A and B of option X, whilst attribute A and B of option Y is discounted. The subtraction allows the drift rate to be positive when evidence from option X is favoured more than option Y, and negative vice versa.

In contrast, the same equation can be constructed for DFT (which describes attribute-wise attention) with some adjustments for the drift rate when comparing between atrributes:

$$(X_A - Y_A) + \phi(X_B - Y_B) \textit{ when looking at attribute } A,$$

$$\phi(X_A - Y_A) + (X_B - Y_B) \textit{ when looking at attribute } B,$$

where ϕ represents the discount factor on the non-fixated attribute ($0 \leq \phi \leq 1$). When attribute A is looked at (regardless of which option), for example, evidence is accumulated from both option X and Y of attribute A, whilst evidence from both option X and Y of attribute B is discounted as a whole.

Merging both DFT and aDDM models together, the model would discount both the non-fixated option (θ) and non-fixated attribute (ϕ). For example, when looking at option X with attribute A (X_A), the drift rate would be:

$$(X_A - \theta Y_A) + \phi(X_B - \theta Y_B), \tag{1}$$

$$X_A - \theta Y_A + \phi X_B - \gamma Y_B, \tag{2}$$

where the non-fixated attribute of the non-fixated option (hereinafter called the "diagonal" element; Yang and Krajbich, 2023, p. 55) is doubly discounted by θϕ in (1) and transformed into $\theta\phi = \gamma$ ($0 \leq \gamma \leq 1$) in (2), as seen in Yang and Krajbich (2023).

Gaze attention and attribute weights are then introduced into the model. The weight parameter for attribute A is assumed to be ω ($0 \leq \omega \leq 1$) whereas the weight parameter for attribute B is assumed to be *1-ω* (in a zero-sum trade-off relationship). An example equation is seen below when looking at attribute A of option X:

$$\omega(X_A - \theta Y_A) + (1 - \omega)(\phi X_B - \gamma Y_B). \tag{3}$$

Furthermore, the symbol for gaze dwell time proportion (as a percentage of total looking time) for each element X_A, Y_A, X_B, Y_B, is represented as p, q, m, and n respectively ($0 \leq p + q + m + n \leq 1$) and fitted into the model as input data. The equation from (3) thus expands into (4) below for the example of looking at attribute A of option X:

$$\omega(pX_A - \theta qY_A) + (1 - \omega)(\phi mX_B - \gamma nY_B). \tag{4}$$

It is important to note that the drift rates described above (1–4) are derived only from looking at attribute A of option X (i.e., X_A). Iterations and adjustments are made when gaze attention shifts to X_B, Y_A, and Y_B. The full model combining all equations (for all elements) are described in the Supplementary Material C.

2.5. Model fitting

In the context of this study, option X and Y represent the two sandwich choices presented in each trial of the food choice task (i.e., X = Left, Y = Right), whilst attribute A and B represent the qualities of Taste (proxy: image of sandwich, value obtained from self-reported Taste Preference) and Sustainability (proxy: carbon label, value obtained from objective Carbon Score) respectively (i.e., A = Taste, B = Sustainability). The parameter ω represents the weight for attribute Taste, and 1-ω for attribute Sustainability.

The analysis followed a computational modelling approach using Bayesian STAN model in R, also known as RSTAN (Carpenter et al., 2017). The models (described in Model Specification) were fitted to data following an adapted and simplified version of Yang and Krajbich's (2023) analysis (see their paper for more advanced model fitting). Whilst Yang & Krajbich's advanced analysis provides

a more in-depth and richer use of the data, we extracted only the main components needed to answer our research question, thus resulting in a simpler analysis. The analysis explanation is split into two paragraphs below for better clarity and ease of understanding, but the analysis was conducted simultaneously in one single step (i.e., the maaDDM was fitted wholly to the data). All R codes, priors used, Gelman-Rubin statistics (R-hat convergence diagnostic), and further specifications (for replication purposes) are provided and explained in the Supplementary Material C.

Firstly, the drift rate (i.e., rate of evidence accumulation) was estimated through a basic DDM framework. Choice (Left = 1, Right = 0) and Reaction Time (RT) data was fitted to the model using the (built-in) RWiener function (Wabersich and Vandekerckhove, 2014) in RSTAN. RT on each trial i follows the Wiener first passage time distribution:

$$RT_i \sim wiener(a, t_{er}, z, \delta_i),$$

where a is the boundary separation, t_{er} is the non-decision time (which includes encoding time and motor response time; see Myers et al., 2022), z is the starting point bias, and δ_i is the trial-level drift rate. This framework can be explained with reference to Fig. 4 (see Note). The drift rate δ_i was estimated for each trial and the model was fitted to each individual subject separately.

Secondly, the drift rate for each trial was re-expressed as the full maaDDM equation (see Supplementary Material C), which consists of a scaling parameter d (i.e., an overall conversion unit to help with model convergence), subjective individual Taste Preferences (i.e., values for X_A and Y_A), objective Carbon Score (i.e., values for X_B and Y_B), attentional discount parameters (i.e., θ, ϕ, and γ), dwell time proportion (i.e., p, q, m, and n), and attribute weights (i.e., ω for attribute A, $1\text{-}\omega$ for attribute B; main focus of the paper). Overall, Taste Preference, Carbon Score and dwell time proportion were inputted as data, and all other parameters were freely estimated (i.e., drift rate, boundary separation, non-decision time, attribute weights, attentional discounts, and scaling parameter; see Supplementary Material C for estimation of within-trial variability via transformation).

3. Results

3.1. Data Cleaning

At trial-level (within each participant), exclusions were applied to trials (i) with less than 0.3s Reaction Time (RT) as this would be too fast for comparisons to be made between options or attributes (indicating invalid trials; see also Yang and Krajbich, 2023), and (ii) with log RT more or less than two standard deviations above individual mean, as they were deemed as either too fast (e.g., accidental pressing without having made comparisons properly) or too slow (e.g., asking questions to the researcher in the middle of a trial, rubbing their eyes or adjusting their glasses) from individual normal. The resulting total number of trials for 78 participants was 7765 trials (median = 100, minimum = 93, maximum = 105 individually).

On a subject-level (across all participants), the exclusion criteria included (i) participants who had less than 80 trials (i.e., ensuring

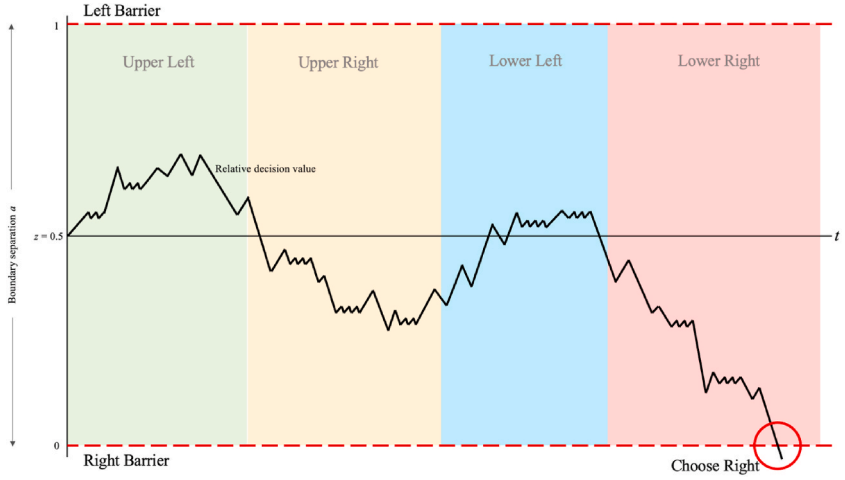


Fig. 4. Illustration of Drift Diffusion Model

Note. The boundary separation a is the distance between the Left Barrier and Right Barrier. As the position of sandwiches were randomised (Left/Right) in each trial, it can be assumed that there should be no significant bias to choose either Left or Right as participants (who have varied preferences for each sandwich) could not possibly predict what they will see or choose. Hence, starting point bias z is fixed at 0.5 (following the study design; see also Yang and Krajbich, 2023) to assume no starting point bias towards the left or right option (i.e., requires same amount of information to reach either side). During decision making, the relative decision value (represented by the noisy black line) fluctuates up and down as a participant samples information from multiple stimuli. When sampling from attributes of the Left option (see green and blue sections), the relative decision value is gradually pushed towards the upper boundary (Left Barrier); when sampling from attributes of the Right option (see yellow and red sections), the relative decision value is gradually pushed towards the lower boundary (Right Barrier). A decision is said to be made when the relative decision value hits a boundary (e.g., choose Right when relative decision value reaches Right Barrier; see Red Circle).

at least 75% valid trials) as this may indicate a lack of proper engagement with the task and may increase uncertainty in model fitting, but no participants were removed due to insufficient valid trials and lack of engagement. Furthermore, we excluded participants who (ii) averaged less than two fixations per trial which may indicate a lack of comparison during task, but no participant was excluded for this reason; (iii) had calibration errors of maximum deviations more than 1.00° (−12 participants); (iv) did not complete the food choice task as required (−1 participant; case28); and (v) had dietary restrictions such as being vegetarian, vegan, or have allergies (−17 participants). This resulted in a final sample of $n = 48$ (62%; $M_{age} = 20.40$, $SD_{age} = 2.26$; female = 32, male = 14, non-binary = 2; British = 24; average fixations = 6.60; average deviation = 0.34°) with a total of 4763 trials (median = 99, minimum = 95, maximum = 104 individually).

3.2. Behavioural data

Descriptive statistics for behavioural data are presented here. On average, participants had a mean RT of 3.10s ($SD = 2.00$), where mean RT did not significantly differ between choosing Left ($M = 3.06$, $SD = 2.02$) and choosing Right ($M = 3.13$, $SD = 1.98$), $t = 1.21$, $p = .227$ (see Fig. 5 for distribution). Meanwhile, dwell time proportion for sandwiches (both left and right combined, average $p + q = 50\ \%$, $SD = 0.17$) was significantly larger than carbon labels (both left and right combined, average $m + n = 13\ \%$, $SD = 0.12$), $t(4762) = 106.04$, $p < .001$, suggesting that participants looked at the sandwiches longer than the carbon labels.

Across all 4 763 valid trials, participants chose the Left option 51.0% of the time. Moreover, 42.8% of all sandwich choices were the tastier and more sustainable option, 27.4% were the tastier but less sustainable option, 25% were the less tasty but more sustainable option, and 4.8% were the less tasty but also less sustainable option. This suggests that participant's choices were generally guided by a combination of taste preferences (slightly more so) and sustainability considerations. However, participants' self-reported taste preferences of sandwiches were not significantly correlated with the objective sustainability ranking of sandwiches, repeated-measures correlation $r(671) < 0.001$, $p = .995$, suggesting no association between taste preference and the environmental impact of sandwiches.

Several regressions were conducted (see Supplementary Material D for full report) to check for the assumptions that make up the maaDDM model (Yang and Krajbich, 2023). We checked for the assumptions of basic DDM, DFT, and aDDM models, and showed that all predictors used in these models were significant, thus adding validity to support the use of maaDDM in the main analysis. We also simulated the choice proportions within each participant using the estimated parameters derived from the maaDDM model below (i.e., boundary separation, non-decision time, and trial-level drift rate), and showed that the model-derived predicted choice proportions (simulation returned 51.0% Left choices) closely aligned with the observed choice proportions (51.0% Left choices), paired-samples t-test = 0.82, $p = .417$.

Furthermore, as a pre-check, the maaDDM model was first simulated in RSTAN without the inclusion of eye-tracking data or any attentional discount parameters, using a logistic regression (outcome = *choice*, predictor = *Taste score difference* and *Sustainbility score difference*). The weights (without gaze attention) attached to Taste and Sustainability were assumed to be ω^* and $1\text{-}\omega^*$ respectively ($0 \leq \omega^* \leq 1$). We tested whether the weights ω^* and $1\text{-}\omega^*$ were significantly different from each other, by checking whether the Bayesian

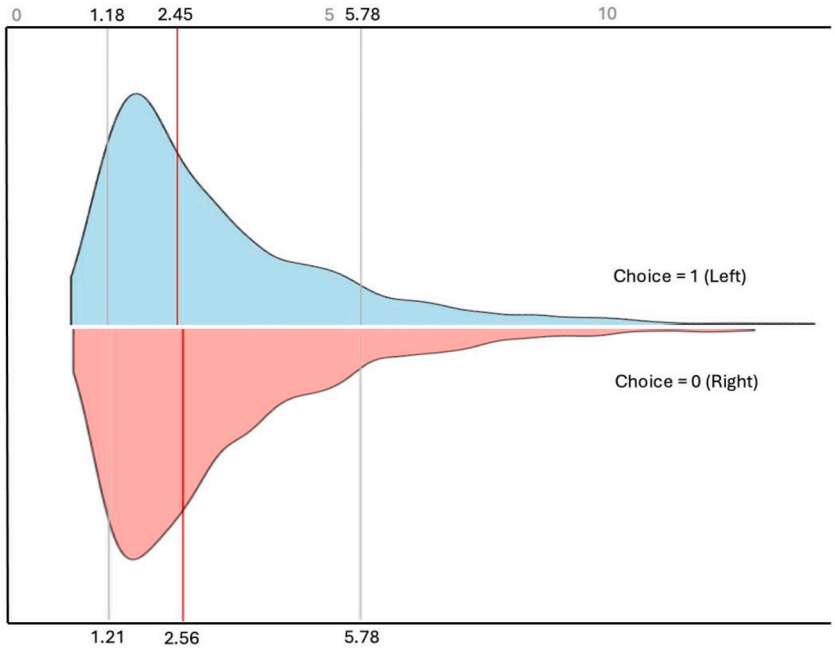


Fig. 5. Distribution of Reaction Time (in seconds) by Choice
Note. Red line denotes 0.5 quantile (median), whilst grey lines to the left and right denote the 0.1 quantile and 0.9 quantile respectively.

Credible Interval (BCI at 95%) from the posterior distribution of ω^* included 0.5. Results showed that the weights for Taste (ω^*) and Sustainability ($1\text{-}\omega^*$) without eye-tracking were not significantly different (mean $\omega^* = 0.66$, BCI = [0.02, 0.99]; Taste-to-Sustainability ratio = 1.94). This is later compared to parameter omega (ω) in the main maaDDM analysis.

Moreover, we have also attempted an econometric approach for model comparison purposes. We ran a mixed logit regression with the "mlogit" package in R (Croissant, 2020) using the same outcome and predictor variable as the pre-check above, except that we assumed the coefficients to be random (i.e., allows for individual-level variability) instead of fixed. In this model, the coefficients attached to Taste and Sustainability were assumed to be β1 and β2 respectively (where β1 and β2 are estimated freely without being constrained in a zero-sum relationship). Results showed that Taste (β1 = 1.75, SE = 0.14, $p < .001$) and Sustainability (β2 = 0.95, SE = 0.08, $p < .001$) were both significant predictors, but the coefficient for Taste (β1) was significantly greater than Sustainability (β2), linear hypothesis test (Chi-squared) = 90.5, $p < .001$. Hence, this suggests that participant's choices were guided by both taste preferences and carbon label information, but taste preferences were almost twice as important than Sustainability in the food decision-making process (Taste-to-Sustainability ratio = 1.94).

3.3. Main analysis: maaDDM outputs

We describe our main analysis here for the model fitting of maaDDM and report all parameter estimates on an aggregated level. Descriptive statistics and Bayesian Credible Intervals (BCI at 95%) for all estimated parameter outputs are summarised across participants (without dietary restrictions) in Table 3 (for individual-level outputs, please refer to dataset "Parameters48.csv" on our OSF page). In contrast with Yang and Krajbich (2023), option-wise discount factor (θ) was larger than attribute-wise discount factor (ϕ), which was larger than diagonal discount factor ($\gamma = \theta\phi$). This demonstrates that, in our sample, the values of options (Left and Right) were discounted less than values of attributes (Taste and Sustainability) when not fixated upon, thus suggesting a trend towards valuing non-fixated options more than non-fixated attributes as competitive information during decision making (cf. Yang and Krajbich, 2023).

To examine whether the attribute weights for Taste (ω) and Sustainability ($1\text{-}\omega$) were significantly different, we checked whether the BCI for ω contained 0.5 (which assumes equal weightage for both attributes). As seen in Table 3, the BCI for Taste attribute weight ω contained 0.5, thus suggesting that, on average across all participants, the weights for Taste and Sustainability did not significantly differ (Taste-to-Sustainability ratio = 1.27). Hence, this may suggest that Sustainability considerations played a moderate role in guiding sandwich decisions; carbon label information was used to a moderate and similar extent relative to individual taste preference during food decision-making.

However, the high uncertainty in parameter estimate should be noted (as seen in the huge range of BCI for ω = [0.09, 0.97]), and we suspect that this is due to the wide range of individual preference for valuing Taste and Sustainability (mean ω minimum = 0.13, maximum = 0.82), as seen in Fig. 6 below. As such, perhaps the aggregated (or group-level) results might not be as informative as desired. Hence, we shifted our focus to examine the weight for Taste (ω) on an individual-level, and explored the heterogeneity of individual preference for Taste/Sustainability considerations.

As seen in Fig. 5, the mean weight for Taste (ω) varied widely between participants. Among all 48 participants, 5 participants had significantly greater weight for Sustainability than Taste (red, BCI below 0.5), 1 participant had significantly greater weight for Taste than Sustainability (blue, BCI above 0.5), but all remaining 42 participants did not differ in their weight for Taste and Sustainability (black, BCI contains 0.5). Overall, this showed some level of variability in individual preference for Taste/Sustainability considerations during food decision-making, which may be more informative than an aggregated preference as we further explore how such preferences vary across individuals by their demographic and climate attitude and perceptions.

3.4. Exploratory analysis: Individual variability

We further explore the variability in individual Sustainability weights ($1\text{-}\omega$) via participant demographic (age and gender) and climate perception and attitudes. We first report the descriptive statistics for all climate perception and attitudes, and check for their correlation with $1\text{-}\omega$ in Table 4.

Table 3
Estimated parameters for maaDDM output.

Parameter	Symbol	*M*	95% CI	*SD*	BCI
Taste attribute weight	ω	0.56	[0.51, 0.62]	0.18	[0.09, 0.97]
Sustainability attribute weight	$1\text{-}\omega$	0.44	[0.38, 0.49]	0.18	–
Option-wise attentional discount	θ	0.73	[0.71, 0.75]	0.08	[0.28, 0.99]
Attribute-wise attentional discount	ϕ	0.43	[0.40, 0.45]	0.08	[0.02, 0.96]
Diagonal attention discount	γ	0.31	[0.29, 0.33]	0.07	[0.01, 0.81]
Scaling parameter	d	0.49	[0.44, 0.54]	0.17	[0.13, 0.95]
Within-trial variability	s	0.38	[0.34, 0.43]	0.16	[0.17, 0.80]
Boundary separation	a	6.09	[5.40, 6.79]	2.39	[2.50, 11.45]
Non-decision time (seconds)	t_{er}	0.33	[0.28, 0.38]	0.18	[0.07, 0.90]

Note. Mean, *SD*, and 95% CI were estimated using $n = 48$ individual posterior mean and SD in a Hierarchical Bayesian model. BCI was generated using $n = 480{,}000$ posterior samples.

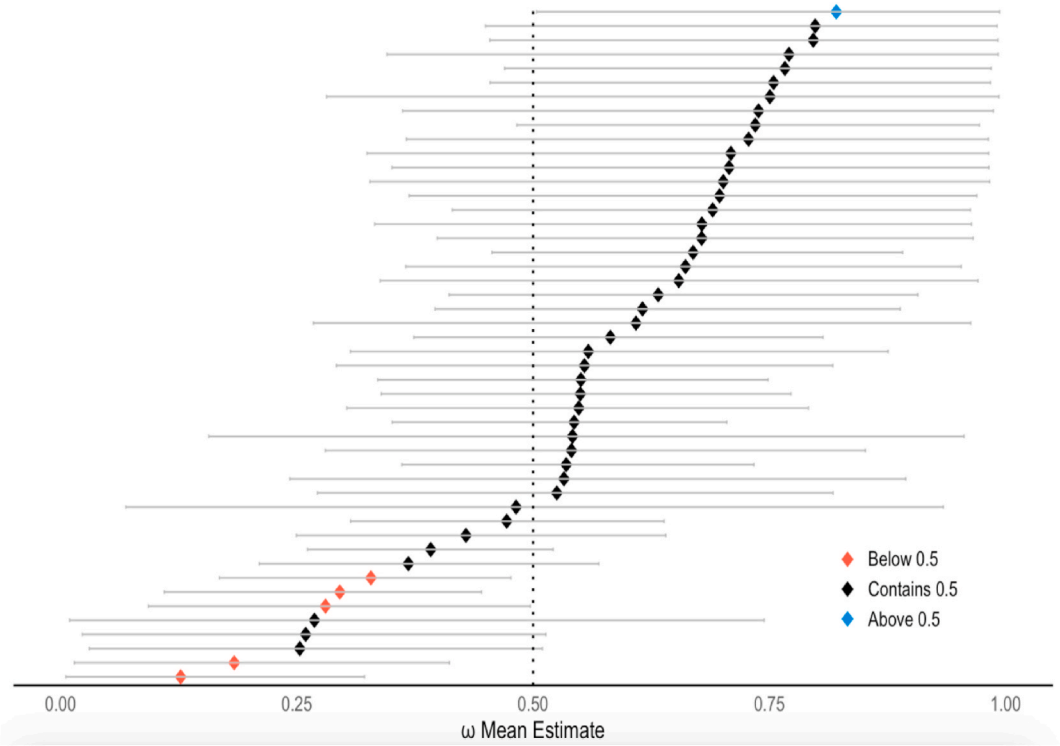


Fig. 6. *Individual-level Mean for Parameter* ω
Note. Whiskers represent BCI at 95% for n = 10,000 posterior samples in each participant.

As seen in Table 4, Pearson's correlational tests showed that Sustainability weight (*1-ω*) was only significantly correlated with NegativeGHG, but not with PolicySupport, Threat, Willingness, and Impact. The individual variables were also significantly correlated (to a medium-to-strong extent) with one another, except for Impact which was not significantly correlated with any of the measures (all r's $\geq$ 0.42, all p's $\leq$ .003; see Supplementary Material E). These variables were then inserted into a hierarchical multiple regression in descending order of correlation magnitude (i.e., largest to smallest) using RSTAN, whilst controlling for participant age and gender (see Howell, 2010). Results are described in Table 5 for each model.

As seen in Table 5, NegativeGHG significantly predicted *1-ω* (BCI does not contain 0) whilst controlling for Age and Gender (Model 1 and Model 2), but it became non-significant once the other variables were added to the regression (Model 3 to Model 5). Threat, PolicySupport, Willingness and Impact were not significant in any of the models. There were also no age nor gender effects. Hence, Model 1 was chosen as the best model that may predict and explain participant's variability in Sustainability attribute weight *1-ω*. This suggests that participants who perceived the effects of GHG emissions and climate change as more negative attached heavier weights to the Sustainability attribute and used carbon labels more in their food decision-making.

4. Discussion

We aimed to empirically measure and quantify the extent to which consumers trade-off between valuing the importance of sustainability considerations versus taste preferences during food decision-making, using a combination of eye-tracking technique and computational modelling approach based on Yang and Krajbich's (2023) Multi-Attribute Attentional Drift Diffusion Model (maaDDM).

Table 4
Descriptive statistics and correlation for climate perception and attitudes.

	Without Diet Restriction (n = 48)		Pearson's r (with 1-ω)	
Variable	*M*	*SD*	*r*	*p*-value
Threat	3.94	0.55	0.26	.073
Willingness	4.79	1.43	0.26	.078
NegativeGHG*	5.97	0.75	0.39**	.006**
Impact	1.73	1.22	−0.14	.333
PolicySupport	4.45	0.61	0.28	.055

**p < .01.

Table 5
Multiple Regression: 1-ω *against Climate Attitudes, Age and Gender.*

	Model 1		Model 2		Model 3		Model 4		Model 5	
Variables	B	BCI	B	BCI	B	*BCI*	B	*BCI*	B	*BCI*
Predictors										
NegativeGHG	0.10[a]	[0.03,0.17]	0.09[a]	[0.004,0.17]	0.09	[-0.02,0.19]	0.09	[-0.02,0.19]	0.09	[-0.01,0.19]
PolicySupport	–	–	0.03	[-0.07,0.13]	0.03	[-0.07,0.13]	0.007	[-0.12,0.13]	0.005	[-0.12,0.13]
Threat	–	–	–	–	−0.003	[-0.12,0.12]	−0.02	[-0.15,0.11]	−0.02	[-0.15,0.11]
Willingness	–	–	–	–	–	–	0.02	[-0.04,0.07]	0.02	[-0.04,0.07]
Impact	–	–	–	–	–	–	–	–	−0.01	[-0.05,0.03]
Covariates										
Age	0.006	[-0.02,0.03]	0.005	[-0.02,0.03]	0.004	[-0.02,0.03]	0.009	[-0.02,0.03]	0.009	[-0.02,0.03]
Gender	0.03	[-0.06,0.12]	0.03	[-0.06,0.13]	0.04	[-0.06,0.13]	0.03	[-0.07,0.13]	0.03	[-0.06,0.13]

Note. Coefficient and BCI computed using Bayesian STAN modelling.
[a] significant (BCI does not contain 0).

In summary, results showed that the weights (or importance) given to Taste and Sustainability did not differ on the aggregate level, but showed some variability on an individual-level. Whilst the majority of participants (87.5%) valued both Taste and Sustainability to a moderate extent on par with one another, some (10%) valued Sustainability more and a minority others (2%) valued Taste more. We further showed that this preference for Sustainability was a function of participant's perception of the negative impact of GHG emissions after controlling for age and gender, such that the more negative perception, the greater carbon label use.

4.1. Taste versus sustainability

The lack of significant differences in weights between Taste and Sustainability in the majority of participants suggest that carbon labels were valued and used to a moderate extent (similar to personal taste preference) in guiding food choices. This is contradictory to previous literature where personal taste preference was often ranked as much more important than sustainability consideration (Embling et al., 2024; Grunert et al., 2014; Roos and Tjarnemo, 2011; van den Berg et al., 2022), and where carbon labels were ineffective and not used in food choices (Babakhani et al., 2020; Dudinskaya et al., 2020; Lin and Nayga Jr, 2022). Instead, the results here add evidence to support the use of carbon labels (Betz et al., 2022; Brunner et al., 2018; Meyerding et al., 2019), although we cannot conclusively claim that they promoted more sustainable choices in our study as we lacked a control group to compare with (i.e., not A/B testing, which we diverged from). Nonetheless, we quantified the extent to which carbon labels were used to inform food choices when they were present which suggested moderate use, although it varied amongst participants (despite everyone having good knowledge on how to use them) as we discuss below.

The exploratory analysis on individual variability of carbon label use further revealed that higher levels of perceived negative impact of GHG emissions was the only predictor for greater weightage of sustainability consideration. As carbon labels are designed to relay information on GHG emissions, the results suggest that participants who perceived the negative impact of GHG emissions to be more serious may have actively searched out the carbon footprint information provided by carbon labels to guide their decision. As such, this may have led to the greater use of carbon labels and thus heavier weightage on sustainability. This is in line with previous studies showing that the use of carbon labels was higher for participants who were more concerned with and aware of the importance of sustainable consumption (Babakhani et al., 2020; Grunert et al., 2014). Taken together, this suggests that raising consumer's understanding, concern and awareness of the impact and importance of sustainable consumption is key to greater use of carbon labels.

However, the other measures of environmental attitudes and perceptions in this study (e.g., willingness to adopt a green diet, and climate policy support) did not significantly predict the weightage for sustainability despite being predictive of sustainable behaviours in other studies (e.g., Capstick et al., 2015; Whitmarsh and Corner, 2017). One explanation for the lack of significant results might be that the different measures may have controlled for one another given their high inter-correlations (see Supplementary Material E). As such, future research should consider exploring wider measures of environmental sustainability, such as environmental values (Maurer and Bogner, 2020) and environmental knowledge (Player et al., 2023), which may predict consumer's use of carbon labels better.

4.2. Multi-attribute Attentional Drift Diffusion Model

The maaDDM also provided some insights into the information sampling process during decision-making. Compared to the parameter estimates (averaged across 7 datasets used) in Yang and Krajbich (2023), we found a higher option-wise discount parameter ($\theta = 0.73$ vs. 0.44) and slightly lower attribute-wise discount parameter ($\phi = 0.43$ vs. 0.56), but very similar diagonal discount parameter ($\gamma = 0.31$ vs. 0.31). This suggests that the participants in our study discounted information less from the non-fixated option and more from the non-fixated attribute, which showed the opposite sampling behaviour compared to the participants in Yang and Krajbich's datasets. However, we are unsure why this was the case; perhaps this is due to the different attributes in comparisons (e.g., two different rewards in monetary outcomes; taste preference vs. nutrition label; clothes preference vs. branding; taste preference vs. carbon labels in our study) which might be more or less attractive to different participants.

Furthermore, the maaDDM outputs can be compared to models without the eye-tracking component. As we showed in our Results section, a logistic regression model without eye-tracking parameters and data generated attribute weight parameters which also did

not differ between Taste and Sustainability, thus suggesting equal weightage when not accounting for gaze attention. However, the outputs from the econometric choice model (mixed logit regression) showed that, whilst both Taste and Sustainability attributes were significant predictors of choice, Taste was twice as important as Sustainability in guiding food choices. Perhaps the discrepancy in results here is due to the different assumptions and methods of examination underlying each model. Whilst we appreciate that an econometric choice modelling approach may have been more straightforward and easily implemented using a simple DCE design online (without the extensive steps to equip eye-tracking in a lab), we argue that perhaps the comparisons between choice models is not a matter of which is better, but which is more suited to answer the research question. As we set out in our study to examine the extent to which carbon label information is sampled and used, we appreciate the suited application of maaDDM that explores the mechanism behind how information is sampled to reach a decision. The maaDDM accounts for not just choice and attribute values, but also reaction time and eye-gaze data to model how information (from attributes) is sampled. This provided a theoretically informed and robust framework for a more nuanced understanding of the process behind decision-making, hence the maaDDM was deemed as a suitable choice model for our research question.

4.3. Implications

The application of the maaDDM (Yang and Krajbich, 2023) in this study could push and expand the sustainability literature to consider more complex questions. For instance, with the ability to quantify the exact valuation of carbon labels and its contribution to food decisions, the next step in this line of research could examine the extent to which carbon labels need to be valued and used to become effective in increasing more sustainable food choices (akin the concept of a critical threshold). The investigation of this critical threshold (i.e., the minimum percentage of valuation needed) could provide crucial insights for policy makers (a) to identify whether the public's level of carbon label use is sufficient to increase more sustainable food choices, and if not, (b) to address the gap in which public's use of carbon label needs to be raised to make the carbon label intervention effective.

Furthermore, obtaining such a critical threshold could help resolve the long-standing debate within the literature regarding whether carbon labels are effective (e.g., Kwasny et al., 2022; Nisa et al., 2019). For example, instead of concluding dichotomously whether carbon labels are effective or ineffective overall, future research could address the mixed results in the literature on the basis of whether the valuation and use of carbon labels surpassed the critical threshold to become effective. This generates further opportunities to examine what factors may help increase the valuation of carbon labels to make them more effective, which helps add more nuance to our understanding of carbon label interventions and the process of human decision-making. This should also provide a clearer and more accurate indication for policy makers to evaluate the success and effectiveness of carbon label intervention, and to better address any gap in ineffectiveness of carbon labels.

4.4. Strengths and limitations

We also acknowledge some limitations in our study. Firstly, the lack of control over other packaging labels (e.g., some has stickers of "High Protein" or "MSC Certified") may pose as a limitation as these other attributes (e.g., health) may have been implicitly inferred but were not accounted for in the model. Although, participants commented informally (post-study) that they did not think about other attributes like price or health during the task, thus suggesting that this limitation may less of a problem. Nonetheless, this study calls for more future studies to explore further how different product attributes (e.g., price, health, branding) may interact and influence the decision-making process, thus giving a more realistic representation of how consumers value and prioritise different product information (Yang and Krajbich, 2023; see proposed expansion of maaDDM to include multiple attributes in Supplementary Material F).

Furthermore, the study did well in addressing the issues of attention and knowledge (Grunert, 2011; Grunert et al., 2014). Firstly, the size of the carbon labels presented on screen was made to be as large and salient as possible so that participants could easily see the labels, thus facilitating attention (Babakhani et al., 2020). Secondly, a knowledge check was conducted before the study to ensure that all participants understood the carbon label system properly and how to use them (Samant and Seo, 2016). This created an ideal experimental setup to examine how much carbon labels are valued when given sufficient attention and knowledge. However, participants may also be primed to consider sustainability more given prior knowledge of, and directed attention towards, the carbon labels. Hence, the sustainability weightage could have been inflated under such ideal conditions, but this may also highlight the conditions required to improve eco-labelling interventions in real-life policies: good education on carbon labels and making them big. Nonetheless, this remains a speculation to be tested experimentally in future studies, where carbon labels can be manipulated to be smaller to represent reality better (i.e., usually taking up a small corner of the food packaging), and prior knowledge on the carbon label system can be withheld from participants. Future research could also examine how well the findings from experimental labs translate into real-life policy effectiveness (see pre-print by Maier and Fesenfeld, 2024).

Lastly, the food choice task may be limited in its design to always show sandwiches options at the top above the carbon labels. As we know that visual attention typically shifts from left to right and top to bottom (for those with left-to-right reading habits; Gobel, 2015; Ryan et al., 2018), participants (mostly UK university students studying in English medium) may have always fixated their attention to the top-left stimuli at the start of each trial. This suggests that participants may have given more visual attention, and hence importance, to stimuli on the top side and left side of the screen. As such, whilst the stimuli were counterbalanced left-to-right between higher and lower sustainable sandwiches, future studies should also counterbalance the top-and-bottom positions between food stimuli and carbon labels to remove any potential effects of positioning.

5. Conclusion

In conclusion, this study presented a novel and in-depth evaluation of carbon labels by quantifying the extent to which consumers valued and used the carbon footprint information provided by carbon labels in their food decisions. Overall, results suggest that carbon labels are used moderately in food decisions when given sufficient attention and knowledge, but this may vary depending on participant's perception of the negative impact of GHG emission. Nonetheless, this supports the use of carbon labels in future policies, whilst providing a useful evaluation framework for future research to further examine how carbon labels could be designed to be more effective.

CRediT authorship contribution statement

Yu Shuang Gan: Resources, Methodology, Formal analysis, Writing – original draft, Visualization, Project administration, Investigation, Data curation, Conceptualization. **Neal Stuart Hinvest:** Supervision, Writing – review & editing, Conceptualization.

Data availability statement

The lead author, Yu Shuang Gan, has full access to the data reported in this study. Furthermore, all data used in this study is available and openly accessible to all via the Open Science Framework (OSF) link online here: https://doi.org/10.17605/OSF.IO/7QHBW.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jocm.2025.100564.

Data availability

Data & Material for Quantifying the Value of Carbon Label Information in Food Choice using Drift Diffusion Modelling (Original data) (Open Science Framework)

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